Kaicheng Xue

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EDUCATION

University of Virginia, Charlottesville, VA Aug 2025 – Dec 2026 (expected)
M.S. in Computer Science, GPA: 4.0/4.0
Thesis: *Markov Dynamics of Hidden States and LLM Reasoning Accuracy* (Advisor: Yangfeng Ji)

University of Wisconsin–Madison, Madison, WI Sept 2021 – May 2025
B.S. in Mathematics, GPA: 3.81/4.0 | B.S. in Computer Science, GPA: 3.78/4.0
Relevant Coursework: Real Analysis, Stochastic Processes, Probability, Abstract Algebra, Linear Optimization, Linear Algebra, Ordinary Differential Equations, Discrete Mathematics
Honors: College of Letters & Science Dean's List

RESEARCH EXPERIENCE

Graduate Research, University of Virginia Jan 2026 – Present
Markov Dynamics of Hidden States in LLM Reasoning (Advisor: Yangfeng Ji)

- Modeled layer-wise hidden-state trajectories of LLMs over chain-of-thought reasoning steps as discrete Markov chains; derived per-layer entropy rates from stationary distributions and characterized mixing behavior via spectral gaps of transition matrices
- Evaluated 11 models from Llama and Qwen families (base, post-trained, LoRA-fine-tuned) across four reasoning benchmarks (HotpotQA, ProofWriter, GSM8K, HSH-Math)
- Designed a rigorous evaluation protocol using nested leave-one-out cross-validation with in-fold feature selection and selection-aware permutation tests
- Found that hidden-state Markov dynamics are confounded with model size/family across different models, but reliably track accuracy changes from fine-tuning within a fixed base model

Student Researcher, University of Virginia Aug 2026 – Present
LLM-Generated Fake-Review Attacks on AI Shopping Agents
PI: Tianhao Wang (CS); Co-PIs: Michael Albert (Darden), Jingjing Li (McIntire)

- Built an end-to-end simulation of an Amazon-style marketplace in which $\sim$200 LLM shopper agents browse, compare, and purchase products per target, to measure how coordinated fake-review attacks shift purchasing behavior at the population level
- Designed the data pipeline from the Amazon Reviews 2023 dataset: matched synthetic personas to real Amazon buyers, calibrated demographics to U.S. BLS Consumer Expenditure Survey data, and constructed 60-product search environments from title-embedding nearest neighbors
- Implemented coordinated LLM-generated fake-review attacks (push, segment-targeted, average-profile imitation) with automated quality checks for length and near-duplicate detection
- Redesigned agent architecture from independent prompts to multi-turn conversations on the Claude API with prompt caching ($\sim$77% cache hit rate), fixed reasoning effort, robust retries, and reproducible run logging
- Ran controlled baseline-vs.-attack experiments across 40 targets and cross-model comparisons across 8 LLMs; observed that attack effects concentrate on targets reaching the first results page, with agents citing fake reviews without suspicion

Course Research Project, University of Virginia Spring 2026
Privacy Attack via Temporal Embedding Inversion on RAG Systems

- Proposed a temporal-aware sliding-window method for inverting conversation embeddings leaked from RAG systems, incorporating dialogue history to improve text reconstruction and sensitive-information extraction
- Constructed a 3,000-conversation dataset with three categories of embedded private information (user profile, financial, medical) and evaluated attack success rates across categories
- Analyzed why a non-linear extension (Temporal Context Fusion Network) fails to improve over the linear sliding-window approach, identifying that non-linear projection distorts embedding-space geometry rather than improving alignment
- Achieved best attack success rates of 41% (user profile) and 53% (medical) with the fine-tuned sliding-window variant

Research Intern, Nanyang Technological University May 2024 – Sept 2024
Internal Mechanisms of LLMs in Sentiment Analysis (Supervisors: Xulang Zhang, Rui Mao, Erik Cambria)

- Analyzed hidden states of a fine-tuned RoBERTa model across all 12 layers and 12 attention heads using K-means, spectral, and hierarchical clustering to probe sentiment encoding
- Discovered that only the final three layers encode sentiment-distinguishing representations, and that 3 of 12 heads in layer 11 are redundant (accuracy $\sim$50% vs. $\sim$79% for others)

- Demonstrated that hidden states encode part-of-speech information in early-to-mid layers, with performance declining in later layers as representations become more task-specific
- Extended analysis to word sense disambiguation and textual entailment, showing that the model's internal mechanisms are task-driven rather than mirroring human reasoning
- Published at ICDMW 2024

PUBLICATIONS

Xue, K., Zhang, X., Mao, R., & Cambria, E. "Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis." In *Proceedings of the 2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2024.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript
Frameworks & Tools: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, MATLAB
Infrastructure: Git, Linux, CUDA, Docker, AWS, LaTeX
Methods: Hidden-state probing, Markov chain analysis, stochastic processes, spectral analysis, clustering (K-means, spectral, hierarchical), benchmark construction, statistical evaluation (bootstrap, permutation tests, cross-validation)

HONORS & AWARDS

College of Letters & Science Dean's List, University of Wisconsin–Madison